INTELLIGENT AYURVEDIC HERBAL PLANT IDENTIFICATION AND ECONOMIC ANALYSIS IN SRI LANKA

COMMON INTEGRATED SOLUTION REPORT

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Report submitted in partial fulfilment of the requirements for the

B.Sc. (Hons) Degree in Information Technology

Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Sri Lanka's Ayurvedic tradition depends on knowledge of about 1,400 medicinal plant species, yet that knowledge is held mainly by elderly practitioners, is scarce in Sinhala, and is hard to connect to the practical decisions of identifying a plant, judging its condition and choosing what to cultivate. This report presents an integrated mobile ecosystem, developed by research group R26-IT-021, that addresses this common problem through four interdependent components. A bilingual Retrieval-Augmented Generation assistant answers English and Sinhala questions from a curated knowledge base, refuses when evidence is weak, and states precautions. A plant identification service uses organ-specific EfficientNetV2 classifiers and a YOLO11 organ detector to name plants from photographs and to show uncertainty. An explainable quality assessment component uses a three-stage EfficientNetV2-B0 pipeline with Grad-CAM to identify the species, assess leaf health and maturity, and show why each result was reported. A cultivation recommendation component helps small-scale farmers choose herbs using soil type, land size, budget and local weather. The components share one Flutter application and one Flask backend and exchange narrow interface contracts. Component evaluation showed that the correct plant entry was retrieved for 5 of 6 English and 6 of 6 Sinhala questions, that all eight audience-interaction tests were routed correctly, that the quality pipeline reached 96.66%, 89.27% and 91.74% held-out accuracy for species, health condition and maturity, and that the identification classifiers reached 91% to 100% on validation. The work also exposed limits: a 20-plant knowledge base, small and partly duplicated image data, and a cultivation dataset too small for dependable model selection. The report concludes that a safety-aware integrated system is technically feasible, and sets out a staged commercialization plan with a freemium model, institutional and farmer-program licenses, a three-year financial projection, and an analysis of intellectual property, compliance and Sustainable Development Goal contributions.

Keywords: Ayurvedic medicinal plants, retrieval-augmented generation, plant identification, explainable AI, cultivation recommendation

ACKNOWLEDGEMENTS

We sincerely thank our supervisor, Mr. Nelum Amarasena, and our co-supervisor, Ms. Kaushika Kahatapitiya, for their guidance, constructive criticism and steady support throughout this research. Their feedback on novelty and measurability shaped the evaluation design of the whole project.

We are grateful to the panel members who reviewed the project at each stage, and in particular for the request that led to the flexible audience design of the knowledge assistant. We thank the Department of Information Technology of the Sri Lanka Institute of Information Technology for the facilities and resources provided, and the students and members of the public who tried the prototype and shared their comments.

We also acknowledge the researchers, dataset contributors and open-source communities whose work supported the technical foundation of the system, and our families and friends for their encouragement during the final year of our degree.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
C1 to C4	Component 1 to Component 4
CSV	Comma-Separated Values
EQ	Evaluation Question
G1 to G7	Group Objectives 1 to 7
Grad-CAM	Gradient-weighted Class Activation Mapping
HKAS	Herbal Knowledge Assistant System
JSON	JavaScript Object Notation
LKR	Sri Lankan Rupee
mAP	mean Average Precision
RAG	Retrieval-Augmented Generation
REST	Representational State Transfer
SDG	Sustainable Development Goal
SP1 to SP4	Subproblems 1 to 4
SUS	System Usability Scale
YOLO	You Only Look Once (object detector)

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

Sri Lanka has one of the richest medicinal plant heritages in South Asia. About 1,400 plant species are commonly cited as being used in traditional medicine [1], and traditional systems still meet an estimated 60 to 70 per cent of the primary health care needs of the rural population [2]. The World Health Organization has repeatedly stressed the need for reliable, safe information on traditional and complementary medicine [3]. The public Ayurveda network alone includes 118 hospitals and 306 clinics, with a further 26 private hospitals and 138 private clinics [4].

Despite this strength, the knowledge chain that runs from the plant in the field to the information in a user's hand is fragile. Knowledge of names, parts used, preparations and cautions is held mainly by elderly practitioners and is poorly digitised, particularly in Sinhala. A national survey of the medicinal material industry identified the absence of systematic cultivation and processing protocols, incorrect identification of plant material and weak quality control as major challenges [2]. Supply is also under pressure: in November 2025 the Sri Lanka Ayurvedic Drugs Corporation reported that registered suppliers could not meet orders for several essential plants, including aloe vera [5]. At the same time, mobile access is widespread, with about 30.3 million mobile connections at the end of 2025 [6], which makes a mobile-first solution realistic.

Our project, **Intelligent Ayurvedic Herbal Plant Identification and Economic Analysis in Sri Lanka** (research group R26-IT-021), responds to this situation with one mobile-first ecosystem. A user can photograph a plant and learn what it is, see an explainable assessment of its condition, ask questions about its traditional uses in English or Sinhala, and, if the user is a farmer, receive cultivation guidance that reflects soil, land, budget and local weather.

1.2 Stakeholders and user needs

We identified five stakeholder groups. Their needs were derived from the literature, from the feedback of the supervisors and the review panel at each project stage, and

from informal comments of students and members of the public who tried the prototype.

Table 1.1 Stakeholders and confirmed needs

Stakeholder	Primary need	Component that responds
General public	Simple, safe, Sinhala or English explanations of common herbs; a way to name an unfamiliar plant	C1, C2, C3
Ayurveda practitioners	Quick professional reference with preparation, precautions and evidence level	C1
Students and researchers	Explained reasoning, scientific names, evidence strength and references	C1, C2
Small-scale herbal farmers	A crop choice that fits soil, plot size, budget and weather; ability to judge plant condition	C3, C4
Herbal dispensaries, manufacturers and institutions	Correct plant identification, consistent quality indicators and dependable raw material supply	C2, C3, C4

The review panel asked specifically that the assistant should not lock a user into a single category and should be able to switch between audience types within a conversation. This request shaped the audience resolution mechanism described in Chapter 3.

1.3 Alignment with the Sustainable Development Goals

The project contributes mainly to SDG 3 (Good Health and Well-being) and SDG 4 (Quality Education), because it gives safer and more accessible information about traditional remedies and preserves knowledge held by elderly practitioners [7]. The cultivation component supports SDG 8 (Decent Work and Economic Growth) by helping small-scale farmers make better-informed planting decisions. Secondary links to SDG 12 (Responsible Consumption and Production) and SDG 15 (Life on Land) are possible through improved plant identification and managed cultivation, but we treat them as potential rather than demonstrated effects. Section 7.5 evaluates these contributions and their limits.

1.4 Purpose and scope of the common report

This report presents the integrated problem and the integrated solution. It explains how the four components divide the problem, how they interlock, how the complete

system was built and evaluated, and how it can be taken to users. The four individual theses provide the detailed component-level research and development evidence, and this report refers to them for experimental detail. The scope covers text and image interaction in a mobile application backed by a Flask service. Clinical diagnosis, treatment advice, production-scale deployment and IoT soil sensing are outside the scope.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

The overall problem is that people who work with Sri Lankan medicinal plants, whether as users, practitioners, students or farmers, lack a single reliable and accessible digital means of moving from an unknown plant to trustworthy knowledge and, where relevant, to an informed cultivation decision. Information is scattered across costly printed books, papers and web pages of uneven quality, is mostly in English, and cannot be searched conversationally. Plant names are frequently confused, so a wrong identification can propagate into wrong use. Farmers must choose among herbs with limited decision support that combines environmental, financial and market factors.

2.2 Root causes and overall research gap

Four root causes can be separated from any proposed technology. First, traditional knowledge is erosion-prone because it is transmitted orally and is concentrated among a shrinking group of practitioners. Second, existing digital tools are polarised: ontologies hold deep knowledge but require query-language skills [8], while chatbots are easy to use but depend on fixed patterns and shallow knowledge [9]. General-purpose language models are fluent but can state unsupported facts, which is dangerous in a health setting [10]. Third, photographic identification tools exist [11], [12], yet they rarely connect an identification to precautions, Sinhala explanations or the condition of the plant. Fourth, agricultural recommendation research concentrates on conventional crops and gives little attention to medicinal herbs or to the budget and land limits of small farmers.

The overarching gap is therefore the absence of an integrated, bilingual and safety-aware system that links identification, quality, knowledge and cultivation decisions for Sri Lankan medicinal plants. No single technique closes it, which is why a four-component solution is required.

2.3 Subproblem and component mapping

We divided the common problem into four non-overlapping but interdependent subproblems. Each is owned by one team member and exposes a narrow interface to the others.

Table 2.1 Subproblems and component mapping

ID	Subproblem	Component and owner	Dependency
SP1	How can users obtain grounded, bilingual and safe knowledge about a plant?	C1 Herbal Knowledge Intelligence System (Munasinghe M P, IT22640598)	Receives plant names and disease labels from C2 and C3
SP2	How can a plant be named from a photograph, with uncertainty made visible?	C2 Plant Identification (H D Sachintha, IT22168986)	Supplies the plant label to the application and to C1
SP3	How can the condition of a plant be assessed and explained visually?	C3 Explainable Quality Assessment (Pattiya Arachchi P A H K, IT22112132)	Supplies disease and quality labels to C1
SP4	Which herb should a small-scale farmer cultivate given soil, land, budget and weather?	C4 Cultivation Recommendation (Wickramasinghe D G T K, IT22132796)	Uses shared application and backend; plant knowledge from the common domain

All four are needed. Identification without knowledge gives a bare label. Knowledge without identification forces the user to already know the plant. Quality assessment without explanation is not trusted, and cultivation advice has little value if the farmer cannot recognise and assess the crop. Together they cover the path from recognition through understanding to production.

2.4 Overall aim and group objectives

Aim. To design, build and evaluate an integrated mobile ecosystem that identifies Sri Lankan Ayurvedic plants, explains their condition and traditional use in English and Sinhala with safety context, and helps small-scale farmers choose suitable herbs to cultivate.

Table 2.2 Group objectives and evaluation evidence

ID	Group objective	Components	Evidence used for evaluation
G1	Provide grounded bilingual answers that carry precautions and refuse when evidence is weak	C1	Bilingual paired question set, interaction suite

G2	Identify supported plants from images and communicate uncertainty	C2	Recorded validation reports, API contract checks
G3	Provide an explainable assessment of plant quality	C3	Component evaluation and Grad-CAM outputs
G4	Recommend suitable herbs from soil, land size, budget and weather	C4	Classification metrics, input and API tests
G5	Deliver the four components through one application with narrow, stable interfaces	All	Interface specification, integration checks
G6	Apply safety, privacy and responsible-AI practice across the system	All	Design review against standards, test cases
G7	Establish a feasible commercialization and sustainability plan	All	Business model, financial projection, risk analysis

2.5 Scope assumptions and constraints

The assistant covers a curated set of 20 plants with precaution and evidence fields. The identification service supports six leaf, ten seed or spice and eight flower classes, and the quality service supports six species: Betel, Centella asiatica, Holy Basil, Kalanchoe pinnata, Murraya koenigii and Neem. The system provides educational and decision-support information only and is not a medical device or a substitute for a practitioner or agronomist. We assume that the external language-model and embedding services are available while the system runs, that users read the educational disclaimer, and that the datasets reflect typical user situations. Constraints are the size of the curated knowledge base, the size of the cultivation dataset, the lack of field-collected specimen-level image data, and the prototype status of deployment, which currently runs on a local server.

2.6 Related work and positioning

Earlier work on Sri Lankan medicinal plants falls into three groups: knowledge representation, image-based identification and conversational access. Each solves part of the problem, and none connects identification, quality, knowledge and cultivation.

Table 2.3 Related work and positioning of this project

Work	Contribution	Limitation for the common problem	How this project responds
Ontology of Sri Lankan medicinal plants [8]	Structured, deep plant knowledge	Query-language skills needed; no images, no Sinhala interface	Natural-language questions in English and Sinhala over a curated base

Ayurvedic knowledge sharing with Sinhala virtual assistant [9]	Sinhala conversational access	Pattern matching with limited phrasing tolerance	Retrieval and query normalisation tolerate varied phrasing
Mobile tool to identify and learn medicinal herbs [13]	Image identification with learning content	Identification only; no condition assessment or cultivation link	Identification joined to quality, knowledge and cultivation
Pl@ntNet and Leafsnap [11], [12]	Large-scale identification	No Ayurvedic precautions, no Sinhala, no condition screening	Domain safety information and Sinhala answers with every plant
General-purpose language assistants [10]	Fluent multilingual answers	Unsupported statements possible in a health setting	Evidence gate, refusal and mandatory precautions

Our positioning is therefore not a better single classifier or a better single chatbot. It is a combination that keeps the user safe at each step and connects steps that were previously separate tools.

2.7 Group-level research questions

Table 2.4 Group-level research questions

ID	Research question	Answered in
RQ1	Can curated, evidence-gated retrieval give safe bilingual answers about Sri Lankan medicinal plants?	Chapters 3 and 6 (C1)
RQ2	Can image models name supported plants and communicate their uncertainty?	Chapters 3 and 6 (C2)
RQ3	Can a staged, species-conditioned model with visual explanation assess leaf health and maturity?	Chapters 3 and 6 (C3)
RQ4	Can a recommender combine soil, land, budget and weather to guide small-scale farmers?	Chapters 3 and 6 (C4)
RQ5	Can the four components be integrated through narrow contracts into one application?	Chapters 4 and 6
RQ6	Is the integrated system commercially and socially sustainable?	Chapter 7

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

The solution is one mobile ecosystem rather than four separate products. A shared Flutter application provides the user interface, and a shared Flask backend exposes the services of the four components through a REST application programming interface (API) with JSON payloads. The central value proposition is that one photograph or one question can lead the user to a named plant, an explainable condition assessment, a grounded and cautious explanation, and, for farmers, a cultivation decision. The end-to-end outcome is a trustworthy path from recognition to understanding to production.

3.2 High level architecture

Figure 3.1 shows the architecture. Users interact only with the Flutter client. The backend hosts the four components and connects them to external language and embedding services, to local data stores and to a location-based weather service. The interfaces between components are deliberately narrow, normally a short label in and a structured result out, so that each component can be developed and tested in parallel.

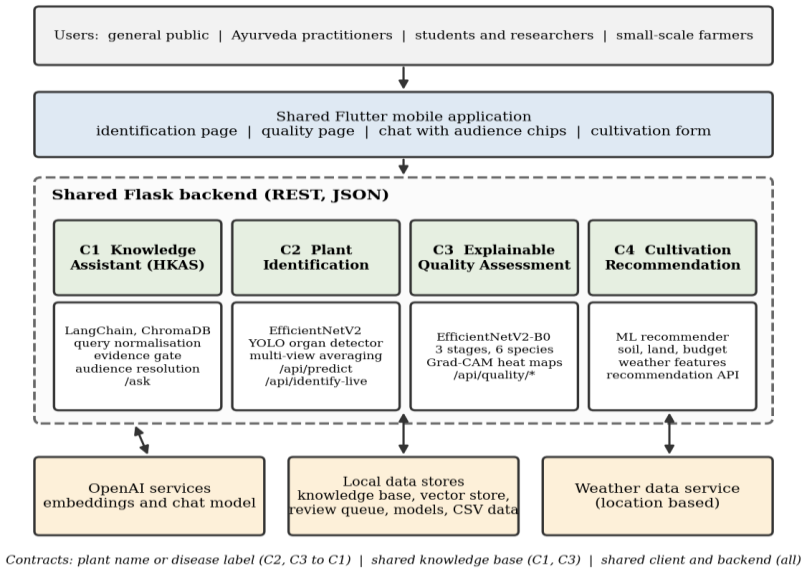


Figure 3.1 High level architecture of the integrated solution

3.3 Component 1 summary: Herbal Knowledge Intelligence System

Subproblem: SP1, grounded bilingual knowledge access. Input: a free-text question in English or Sinhala, a session identifier and an audience context, or a short plant name or disease label from another component. Core method: the Herbal Knowledge Assistant System (HKAS), a Retrieval-Augmented Generation (RAG) pipeline over a curated knowledge base of 20 plant entries and plant-disease entries. Each entry is embedded as one chunk with text-embedding-3-large and stored in ChromaDB. A single language-model call normalises the query by resolving pronouns from history, translating Sinhala into English and splitting questions about several plants. An evidence gate discards retrieved chunks beyond a cosine distance of 0.8 and, for Sinhala, applies a lexical guard. Four audience categories (general public, practitioner, student, researcher) can be chosen explicitly, detected per message, or combined. Output: an answer in the language of the question with precautions and evidence level, the sources used, and safety flags. Contribution: it is the explanatory and educational layer that turns a label into usable and safe information.

3.4 Component 2 summary: Plant Identification

Subproblem: SP2, naming a plant from images. Input: up to five photographs and a selected category (leaf, seed or spice, flower), or one live camera frame. Core method: separate EfficientNetV2 classifiers for each category, image-quality heuristics, averaging of eight views (four crop scales and their mirrors) per image and then across images, and a YOLO11 organ detector that routes live frames to the right classifier. Output: the predicted label, ranked alternatives, a confidence warning below 0.95, image-quality feedback, and educational plant information. A separate backend workflow stores requests for botanical expert review. Contribution: it provides the entry point of the ecosystem and supplies the plant label that downstream components explain.

3.5 Component 3 summary: Explainable Quality Assessment

Subproblem: SP3, assessing and explaining the condition of a medicinal leaf. Input: one to three leaf images, with optional leaf length and width measured by the user. Core method: a hierarchical three-stage pipeline built on EfficientNetV2-B0

backbones [14]. Stage 1 identifies one of six supported species and rejects other inputs through an Unknown class and a 0.60 confidence gate. Stage 2 uses a shared backbone with six species-specific output heads to classify 36 visible health conditions. Stage 3 classifies maturity (young, developing or mature) only for leaves found healthy in Stage 2. A pre-inference check rejects images that are too small, blurred or badly lit, and the probabilities from up to three views are averaged. Gradient-weighted Class Activation Mapping (Grad-CAM) [15] highlights the leaf regions that drive the prediction. Structured knowledge files translate each result into a description, harvesting advice and a medicinal suitability level (suitable, conditional or unsuitable), and a decision layer combines the model's maturity result with the user's manual measurements. Output: species, condition, maturity, suitability and a Grad-CAM image in one response. Contribution: it adds trust and transparency by showing the evidence behind a quality judgement, and it lets a user continue from a diseased leaf to the knowledge assistant for further guidance.

3.6 Component 4 summary: Cultivation Recommendation

Subproblem: SP4, selecting herbs to cultivate. Input: soil type, available land size and cultivation budget in Sri Lankan rupees (LKR), together with weather information for the farmer's location. Core method: a machine learning recommender trained on a cultivation dataset of environmental, requirement and economic features. Candidate models are compared on accuracy, precision, recall and F1-score. Inputs are validated, and negative budgets are rejected. Output: recommended herbal plants displayed in the Flutter application. Contribution: it adds the economic and production dimension of the project, so that recognition and knowledge can lead to livelihood decisions.

Table 3.1 Comparison of the four components

Aspect	C1 Knowledge	C2 Identification	C3 Quality	C4 Cultivation
Data type	Text	Image	Image	Tabular
Core technique	RAG with evidence gate	EfficientNetV2 and YOLO11	Three-stage EfficientNetV2-B0 with Grad-CAM	Machine learning recommender
User output	Grounded bilingual answer	Label, alternatives, warning	Species, condition, maturity and	Ranked herb suggestions

			heat map	
Safety mechanism	Refusal, mandatory precautions	Confidence warning, expert review	Unknown rejection, image validation, visual explanation	Input validation, advisory wording
Main interface	/ask	/api/predict, /api/identify-live	/api/quality/*	Recommendation route

3.7 End to end user and data flow

Figure 3.2 shows a typical journey. A user photographs a plant. Component 2 proposes a name and shows its uncertainty. Component 3 checks the image, confirms the species, assesses the condition and, through Grad-CAM, shows where the assessment comes from. For a diseased leaf the user can continue to Component 1 with the detected species and condition, which is expanded into a request for what the disease is, its typical symptoms and whether affected leaves can still be used. The user can also ask a free-text question in either language at any time. A farmer can then use Component 4 to decide what to grow. A safety layer runs through the whole flow: confidence warnings, evidence gating and refusal, precautions with every use, an educational disclaimer and the expert-review route.

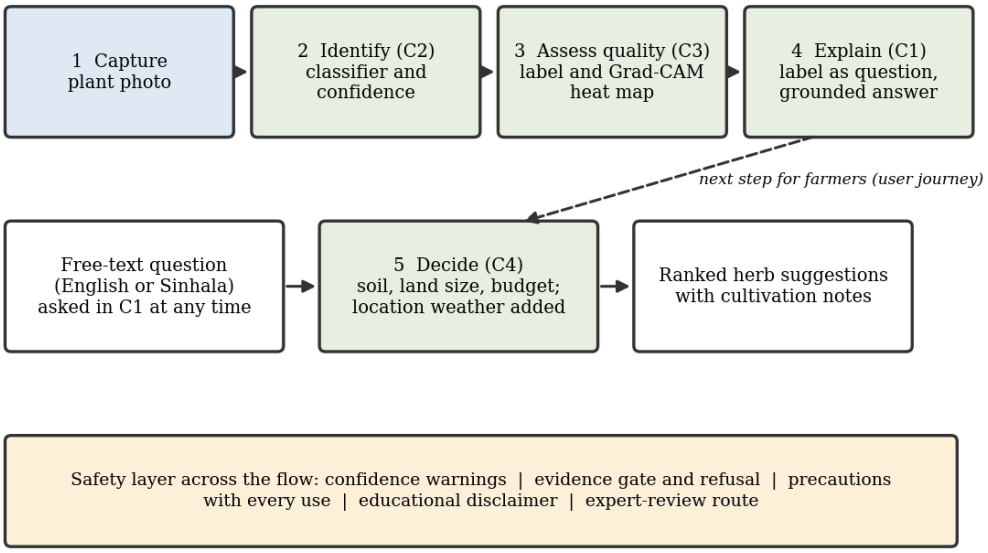


Figure 3.2 End to end user and data flow

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

We followed an interface-first strategy. The team agreed the shape of each request and response before implementation, so that four people could work in parallel on a single repository. Components share one Flask application with separate route groups and one Flutter application with separate pages. Integration risks, mainly mismatched field names, different upload field names and unavailable models, were managed by keeping contracts small, by validating inputs at the boundary, and by returning structured error responses instead of failing silently. The environments used were local development machines for each member and a shared local backend for combined testing.

4.2 Interfaces data contracts and shared models

Table 4.1 summarizes the principal contracts. The full specification is in Appendix A. Two points matter for integration. Prediction and live scanning use different upload field names and must not be treated as interchangeable. The assistant treats an English input of at most six words without a question mark as a label sent by another component and expands it into a full question.

Table 4.1 Principal interface contracts

Interface	Direction	Payload	Notes
/ask	Client to C1	Question, session id, user context (home category, explicit category)	Returns answer, sources, categories used, safety flags
/api/predict	Client to C2	Repeated images, model_type (plant, seed, flower), optional category validation	Warning below 0.95; 400 and 503 errors; 25 MiB limit
/api/identify-live	Client to C2	One frame	No detection gives a null identification
Short label	C2 or C3 to C1	Plant name or disease label as a short question	Expanded into a full question by C1
/api/quality/validate-image, identify-plant-multiple, assess-condition	Client to C3	One to three leaf images; optional manual leaf measurements	Species, condition, maturity, suitability and Grad-CAM image; 422 for rejected images
Recommendation route	Client to C4	Soil type, land size, budget, location	Recommended herbs

4.3 Shared nonfunctional requirements

Performance and reliability. The backend degrades to a polite message when retrieval or the language model fails, distinguishes upload failures from unavailable models, and enforces a request-size limit. **Usability and accessibility.** The interface is simple enough for all age groups, offers bilingual text, and uses clear colour contrast. **Security.** Secret keys are held in environment variables, inputs are validated and bounded, and the design follows OWASP guidance for language-model applications [16]. **Privacy.** The assistant does not collect personal data. The design is consistent with Sri Lanka's Personal Data Protection Act [17]. **Maintainability and scalability.** Knowledge is updated by editing text files and re-ingesting, models are loaded lazily and cached, and the component structure follows the quality characteristics of ISO/IEC 25010 [18]. Production hardening, such as restricted cross-origin settings, HTTPS and rate limits, is part of the commercialization plan.

4.4 Integrated workflow and exception handling

In normal operation the client calls the relevant route, the backend validates the request, the component produces a structured result and the client renders it. Important failure paths are handled as follows. An image that is too small, blurred or badly lit is rejected by the quality service before any model runs, and an unsupported species is routed to the Unknown class instead of receiving a condition. Maturity is assessed only for leaves found healthy. A confident mismatch between the chosen image category and the detected organ returns a category-mismatch error with a suggested category. A low classifier score still returns a label but with a visible warning, and no unknown-species output exists yet. If no plant organ is detected in a live frame, the identification is null rather than forced. If the assistant finds no evidence close enough to the question, it refuses in the language of the question and returns no sources. If a model file or the language service is unavailable, the backend returns a service-unavailable response, and the client shows an understandable message. Invalid cultivation inputs are rejected before they reach the model.

4.5 Team contribution and dependency matrix

Table 4.2 Team contribution and dependency matrix

Member	Research and development tasks	Integration work	Test evidence
Munasinghe M P (C1)	Knowledge base of 20 plants with precaution and evidence fields; RAG engine; query normalization; evidence gate; audience resolution; Flask assistant API; Flutter chat and persona pages	Short-label handling for plant names and disease labels; shared disease entries for C3	Bilingual paired set (12 queries); interaction suite (8 tests); evaluation scripts
H D Sachintha (C2)	Classifier training and export; image preprocessing; multi-view prediction; YOLO organ detector; expert-review backend; Flutter identification page	Upload and live-scan contracts with the shared client and backend	Saved validation reports; dataset audit; API contract checks
Pattiya Arachchi P A H K (C3)	Three-stage EfficientNetV2-B0 pipeline; image validator; species-specific health and maturity heads; Grad-CAM; knowledge lookups; manual evidence fusion; Flutter quality page	Species and condition context handed to C1; quality routes in the shared backend	15 unit tests; stage metrics on held-out sets; latency measurements
Wickramasinghe D G T K (C4)	Cultivation dataset; feature design; model comparison; recommendation API; Flutter input form with validation	Recommendation route in the shared backend; weather feature retrieval	Input, boundary, API and integration tests; model metrics

4.6 Security and privacy design

Security was treated as a design requirement of every component and not as a final check. The controls below apply to the shared backend and are described against the threats they address.

Table 4.3 Security and privacy controls

Threat	Control in the design	Component
Prompt injection or manipulated questions	Answers are generated only from retrieved knowledge entries; the evidence gate discards weakly related text and the assistant refuses when no entry is close enough	C1
Exposure of keys and credentials	Service keys are supplied through environment variables and are kept out of the repository; expert endpoints require a key header	All, C2
Malicious or oversized uploads	Total request size is limited to 25 MiB, file types and image counts are validated, and invalid requests return JSON error bodies	C2, C3
Unreadable or unsuitable images	The pre-inference validator checks resolution, blur and brightness and returns a reason code before any model runs	C3
Out-of-scope inputs treated	An Unknown class and a confidence gate reject	C2, C3

as supported	unsupported species; low classifier scores show a visible warning	
Personal data retention	The assistant collects no personal data; review submissions require consent and strip image metadata	C1, C2
Invalid or harmful numeric inputs	Cultivation inputs are validated at the form and the API, and negative budgets are rejected	C4

4.7 A combined session step by step

A combined session illustrates how the contracts work in practice. A herbal collector opens the application and selects the quality page. The client sends one to three leaf images to the validation route. If the images pass, the identification route returns a species and a confidence value, or an Unknown result that stops further analysis. The assessment route then returns the health condition, the maturity stage for healthy leaves, a suitability level, and a Grad-CAM image in a single response. If the leaf is diseased, the collector moves to the assistant page, where the detected species and condition become a question. The assistant retrieves the matching plant and disease entries, checks that the evidence is close enough, and answers in the language of the question with precautions. Later the collector, or a farmer in the same community, opens the cultivation form, enters soil type, land size and budget, and receives a ranked list of herbs. At no point does the user need to know which component produced which part of the answer, which is the intended effect of the shared application and narrow contracts.

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

Python is the common backend language. Flask provides the REST service, and Flutter with Dart provides the mobile client, which lets all four components share one application shell. Table 5.1 lists the technologies and the reason for each major choice.

Table 5.1 Common and component technologies

Layer	Technology	Purpose
Backend	Python 3, Flask	Lightweight REST service shared by all components
Client	Flutter and Dart	One cross-platform mobile application
Retrieval and generation (C1)	LangChain, ChromaDB, text-embedding-3-large, gpt-4o-mini [19]	Entry-level retrieval, cosine-distance gate, low-cost answers
Image models (C2)	TensorFlow and Keras, EfficientNetV2, YOLO11 [14], [20]	Category classifiers and organ detection
Image models (C3)	TensorFlow and Keras, EfficientNetV2-B0 multi-head models, Grad-CAM, CSV knowledge files	Species, condition and maturity classification with visual explanation
Recommendation (C4)	scikit-learn style models, CSV dataset, weather service	Tabular recommendation using farmer and weather features
Persistence	Text files, persisted vector store, SQLite, CSV	Knowledge, embeddings, expert-review queue, cultivation data

The main design choices were: one chunk per knowledge entry so that a use is never separated from its precaution; a distance threshold of 0.8 that is inspectable and tunable; separate classifiers per plant organ so that class sets are explicit; and a hosted language model to keep cost and latency low for a prototype.

5.2 Development and integration process

Work was organized in iterative cycles aligned to the supervisor and panel reviews at proposal, progress and final stages. All source code was kept under version control, secrets were kept out of the repository, and each member developed on a branch before merging. The knowledge component followed a six-stage procedure of

baseline, enrichment, gating and language handling, flexible audiences, evaluation and refinement. Feedback from the panel at an interim review led directly to the four-category audience scheme. Defects found during integration, such as mismatched upload field names and prompt-label leaks, were recorded and resolved in later cycles.

5.3 Deployment and operational configuration

The prototype runs locally. The Flask service starts on port 5000 with the language-service key supplied through an environment variable, the persisted vector store is loaded or rebuilt from the knowledge files, and the Flutter client connects to the configured server address. The backend limits total request size to 25 MiB and answers oversized, missing or wrongly routed requests with JSON error bodies. For production we plan HTTPS on a cloud host, restricted cross-origin settings, rate limiting, logging, protected credentials for expert reviewers, and backup of the knowledge base, model files and review database. Section 7.3 gives the operating cost of this configuration.

5.4 Evidence of team execution

Execution evidence consists of the shared repository history, the iterative changes recorded after each panel review, the scripted evaluation outputs and the contribution declaration in Appendix B. The repository shows the separation of component code from shared route and client modules, which supports the division of work in Table 4.2.

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

The group-level evaluation asks four questions. Does each component deliver its specified function (EQ1)? Are the interfaces between components consistent (EQ2)? Does the system behave safely when evidence or confidence is weak (EQ3)? Is there evidence of usefulness to stakeholders (EQ4)? For C1 we used retrieval hit, wrong refusal, language correctness, precaution presence, category pass and keyword pass. For C2 we used accuracy and macro F1 on recorded validation splits, for C3 stage accuracy, macro F1, latency and unit tests and detector precision, recall and mean average precision (map). For C4 we used accuracy, precision, recall and F1-score. Because the evaluation sets are small, we report counts and inspect cases and do not claim statistical significance.

6.2 Integration and end to end testing

Interface-level checks confirmed that the Flutter service and the Flask routes agree on multipart images, category names and result fields for identification, and that the assistant accepts short labels and expands them correctly. Input validation, boundary and API tests were designed for the cultivation route, including rejection of negative budgets. Security and privacy were checked by design review: keys are outside source control, inputs are bounded, and no personal data is stored by the assistant. The quality service responds in about 0.5 to 0.7 seconds for one image and 1.2 to 1.8 seconds for three, well inside the client time-outs. A complete multi-device run across all four components, and concurrency measurements for the whole system, are listed as continuation work in Section 8.3.

6.3 Overall results

Table 6.1 Synthesized component results

Component	Principal result	Reading of the result
C1 Knowledge	Correct plant entry retrieved for 5 of 6 English and 6 of 6 Sinhala questions; no answerable question refused; 5 of 6 Sinhala answers written in Sinhala; 8 of 8 interaction tests routed to the expected audience categories	Grounded answering works in both languages; Sinhala phrasing, a plant and disease overlap in retrieval and prompt-label leaks need refinement

C2 Identification	Recorded validation: 14 of 15 (93.33%) leaf, 59 of 59 seed or spice, 32 of 35 (91.43%) flower; detector mAP50 0.6421 on 147 validation images	Results are development-stage validation on small splits, with exact duplicates found in the leaf and flower collections; field accuracy is not yet established
C3 Quality	Held-out accuracy of 96.66% for species (with 100% precision on Unknown rejection), 89.27% for 36 health conditions (macro F1 91.22%) and 91.74% for maturity; 15 of 15 unit tests passed; about 0.5 to 0.7 s per image	All three stages perform well on held-out sets; the Betel health head and the Murraya maturity head are the weakest, and results are for six species
C4 Cultivation	Interface, validation and API delivered; first model comparisons on the current dataset gave low scores for every candidate	The dataset is too small for dependable model selection; expansion is the priority

Two cross-cutting findings emerged. First, every component produces a result that is useful but bounded, so the system must always carry uncertainty forward: a plant label with its confidence, a condition with its heat map, an answer with its precaution and evidence level. Second, the quality of the shared knowledge and of each dataset limits the system more than the model architecture does, which motivates the data and expert-review investment in the commercialization plan.

6.4 Stakeholder and user validation

During development the working chatbot was used informally by students and members of the public, whose spoken comments guided interface behavior and the clarity of answers. No formal participant study and no review by Ayurvedic practitioners or botanists has been completed. A planned study will ask users to capture images, choose a category, interpret warnings and find the disclaimer, measuring task completion, errors and the System Usability Scale (SUS). Botanical reviewers will check identification labels against a curated image set, and practitioners will review the knowledge base. These activities are scheduled in Phase 0 of the roadmap in Chapter 7.

6.5 Achievement of common objectives

Table 6.2 Achievement of group objectives

Objective	Status	Justification
G1 Grounded bilingual answers	Largely achieved	Evidence gate, precautions and bilingual retrieval work; Sinhala phrasing and out-of-base tests remain
G2 Image identification	Partly achieved	Working pipeline and warnings;

with uncertainty		independent field testing and calibration are pending
G3 Explainable quality assessment	Achieved for six species	Three-stage pipeline with Grad-CAM, image validation and knowledge lookups met its stage targets
G4 Cultivation recommendation	Partly achieved	Interface, validation and API delivered; model performance needs a larger dataset
G5 Integration through one application	Partly achieved	Contract-level integration achieved; complete device-level run pending
G6 Safety, privacy and responsible AI	Largely achieved	Refusal, mandatory precautions, disclaimers, no stored personal data; expert validation pending
G7 Commercialisation plan	Achieved	Chapter 7 gives the business model, financial projection, risks and compliance plan

6.6 Integrated limitations and threats

The knowledge base covers 20 plants against roughly 1,400 in use, so real users would meet many refusals. Model calls are not deterministic, and each scripted question ran once, so variation between runs is unknown. Validation data for the image models are limited and contain exact duplicates across splits, specimen-level split records are missing, and confidence scores are not calibrated [21]. The evaluation sets are too small to generalize, no out-of-knowledge-base negative set has yet been run, and fact mixing between plants cannot be detected by keyword checks. The cultivation dataset is small and may not represent every Sri Lankan growing zone, and market demand changes over time. The system depends on proprietary language services that can change, cost money and receive user queries. Finally, the components have been integrated at contract level, and a full end-to-end demonstration across all four remains to be recorded. These limitations determine the order of the work in Phase 0 of the commercialization roadmap.

6.7 Integrated scenarios used to check cohesive behavior

Table 6.3 lists the scenarios used to check that the components behave coherently when used together. They are written from the interface contracts and the safety rules of Chapter 4, and each one states the behavior the system must show.

Table 6.3 Integrated scenarios and expected behaviour

No.	Scenario	Expected behaviour	Components

S1	User asks a Sinhala question about a plant that is in the knowledge base	Answer in Sinhala with precautions and evidence level; sources returned	C1
S2	User asks about a plant that is not in the knowledge base	Polite refusal in the question language; no sources returned	C1
S3	User uploads a sharp, well-lit image of a supported leaf	Species, condition, maturity and heat map returned in one response	C2, C3
S4	User uploads a blurred or very dark image	Rejected with a reason code before any model runs	C3
S5	User uploads a leaf of an unsupported plant	Unknown result; no condition is assigned	C2, C3
S6	Classifier confidence is below the warning level	Label shown with a visible warning and ranked alternatives	C2
S7	Leaf found diseased and user continues to the assistant	Question built from species and condition; grounded answer with precautions	C3, C1
S8	User selects a category that does not match the detected organ	Category-mismatch error with a suggested category	C2
S9	Farmer enters a negative budget	Input rejected before the model is called	C4
S10	Language or model service is unavailable	Service-unavailable response and a clear message in the client	All

Scenarios S1 to S2 and S9 were exercised through the assistant interaction suite and the cultivation validation tests. Scenarios S3 to S6 follow the behavior measured in the quality and identification services, and S7 follows the context hand-off that links the quality page to the assistant. S8 and S10 follow the error contracts in Appendix A. A recorded run of all ten scenarios on a single device is the first item in the continuation work of Section 8.3.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

This chapter sets out how the integrated system can be taken from a research prototype to a sustainable product. We use the working name **HerbIntel Lanka** for the commercial offering. The plan is staged: first validate and harden the prototype, then release it publicly on a freemium basis, then scale through institutional and farmer-programme licenses. All figures labelled as planning assumptions are targets to be tested in the pilots and are not measurements.

7.1 Value proposition and beneficiaries

Value proposition. HerbIntel Lanka gives users a bilingual, safety-aware route from an unknown plant to a named species, an explained condition, a grounded explanation and, for farmers, a cultivation decision, in one mobile application. Unlike a general chatbot it answers only from a curated source and declines when evidence is weak. Unlike a plant-identification app it connects a name to Sinhala explanations, precautions and cultivation guidance.

Table 7.1 Beneficiaries and value created

Beneficiary	Pain point	Value delivered	Value type
General public	Cannot name or safely use common herbs; little Sinhala material	Free bilingual explanations with precautions and identification	Safety, access
Practitioners	Time lost searching references; knowledge held in memory	Fast reference with evidence levels and preparation notes	Time saving, consistency
Students and researchers	Scattered sources, no evidence levels	Explained answers, scientific names, evidence strength	Learning, research aid
Small-scale farmers	Uncertain crop choice, limited budget, weather risk	Recommendations fitted to soil, land, budget and weather; condition checks	Income, reduced risk
Dispensaries, manufacturers, exporters	Misidentified material, inconsistent quality, unreliable supply	Identification and quality screening support; access to cultivation program	Quality, supply security
Society and heritage	Loss of knowledge held by elderly practitioners	Searchable bilingual record of traditional knowledge	Cultural preservation

Stakeholder evidence is currently developmental: panel requests shaped the audience scheme, and informal users commented on answers and interface. Section 6.4 describes the formal validation that must precede paid deployment.

7.2 Market alternatives and differentiation

7.2.1 Market context

Demand for reliable herbal information and raw material is visible. Traditional medicine remains central to rural primary care [2], the public and private Ayurveda network comprises 144 hospitals and 444 clinics [4], and shortages of supply have been publicly reported [5]. On the digital side, Sri Lanka had about 30.3 million mobile connections at the end of 2025, 91.3 per cent of them broadband [6], and about 13.9 million internet users, although roughly 40 per cent of the population remains offline [22]. These figures support a mobile-first strategy but also show that the offline population needs institution-mediated reach, such as clinics and cooperatives.

Table 7.2 Target segments and entry sequence

Segment	Indicator of scale	Entry route	Priority
Students and practitioners	Ayurveda hospitals and clinics [4]; Ayurveda education institutions	Institutional pilot, then paid Plus plan	1
General public	Internet users of about 13.9 million [22]	Free tier through app stores and social media	1
Farmer cooperatives and agri programs	Programs promoting herbal cultivation	Program license with farmer onboarding	2
Manufacturers and exporters	Supply shortages reported [5]	API and quality-screening license	3

7.2.2 Alternatives and differentiation

Table 7.3 Alternatives compared with the integrated product

Alternative	Strength	Gap that HerbIntel Lanka addresses
General-purpose chat assistants	Fluent, multilingual, free	Not grounded in a curated Sri Lankan source; may state unsupported health facts [10]; no refusal tied to evidence
Plant identification apps [11], [12]	Strong image recognition and community data	No Ayurvedic preparation, precautions or Sinhala knowledge; no cultivation link
Sinhala herbal chatbots and ontologies [8], [9]	Local language, deep structured data	Pattern matching or query-language skills; limited phrasing tolerance; no images
Weather and agricultural advisory apps	Environmental data, general crop advice	Not tailored to medicinal herbs or to a farmer's budget and plot size
Printed references and practitioner consultation	Trusted, expert-led	Costly, not searchable, limited availability, mostly English

Our differentiation rests on five features: evidence-gated answers that refuse rather than guess; mandatory precautions and evidence levels; Sinhala support through query normalization; audience-adaptive answers; and one integrated path from identification to cultivation. We are candid that the underlying model families can be copied. The defensible assets are the curated and expert-reviewed bilingual knowledge base, the evaluation data and error taxonomy, trusted institutional partnerships, and the integrated workflow.

7.2.3 Representative user journeys

Four short journeys show the value in use and guided the feature priorities of the plan.

Table 7.4 User journeys and value

User	Situation	Journey through the product	Result
Ayurveda student	Preparing for an examination on Neem and Tulsi	Asks in Sinhala, switches to student mode, opens evidence levels and references	Faster revision with traceable sources
Herbal collector	Unsure whether harvested leaves are fit for use	Photographs leaves, receives species, condition, maturity and suitability with a heat map	Informed decision before sale or processing
Small-scale farmer	Has 0.5 acre, limited budget and clay-rich soil	Enters soil, land and budget, receives ranked herbs with local weather context	Crop choice matched to resources
Dispensary owner	Receives mixed leaf lots from suppliers	Uses identification and quality screening on sample leaves, reviews flagged items with an expert	Fewer unsuitable lots accepted

7.3 Business model and feasibility

7.3.1 Business model summary

Table 7.5 Business model summary

Element	Plan
Customer segments	Public, practitioners and students, farmer programmes, institutions and manufacturers
Value proposition	Safe, bilingual, integrated herbal knowledge, identification and cultivation guidance
Channels	App stores, Ayurveda colleges and clinics, cooperatives, extension programmes, social media
Customer relationships	Free self-service, institutional onboarding and training, expert-reviewed content updates
Revenue streams	Plus subscriptions, institutional licences, farmer-programme licences, grants and pilot funding

Key resources	Curated knowledge base, trained models, evaluation toolkit, mobile and backend code, expert advisers
Key activities	Content curation and review, model maintenance, field testing, partner support, compliance
Key partners	Department of Ayurveda, Ayurveda education institutions, practitioners, cooperatives, universities, incubators
Cost structure	Staff, cloud and model usage, content review, marketing and pilots, legal and compliance

7.3.2 Pricing and revenue streams

The model is freemium. Core safety information and identification stay free so that the public benefit is not restricted. Revenue comes from users and organisations that need depth, volume or integration.

Table 7.6 Proposed plans and pricing (planning assumptions)

Plan	Audience	Features	Price (LKR)
Community	General public	Bilingual questions with daily limit, identification, condition check	Free
Plus	Practitioners, students, researchers	Unlimited questions, practitioner and researcher modes, evidence references, saved history	590 per month or 6,000 per year
Institution	Clinics, colleges, dispensaries, manufacturers	Multi-user licence, custom knowledge entries, usage reports, support	180,000 per year
Farmer programme	Cooperatives, NGOs, agri programmes	Farmer onboarding for up to about 200 farmers, cultivation recommendations, training	250,000 per year; farmers free

7.3.3 Go-to-market and roadmap

Figure 7.1 shows three phases. Phase 0 (months 0 to 6) validates and hardens the system: expert review of the knowledge base, a specimen-level field test set, Sinhala and gate test suites, resolution of dataset and licence records, and pilots with one college and one cooperative. Phase 1 (months 7 to 18) is a public Android beta with the freemium plans and institutional pilots, with the knowledge base grown to about 60 plants and a usability study. Phase 2 (months 19 to 36) scales through farmer-program licenses, a manufacturer API, a Tamil interface, an iOS client and a larger knowledge base with regional cultivation data.

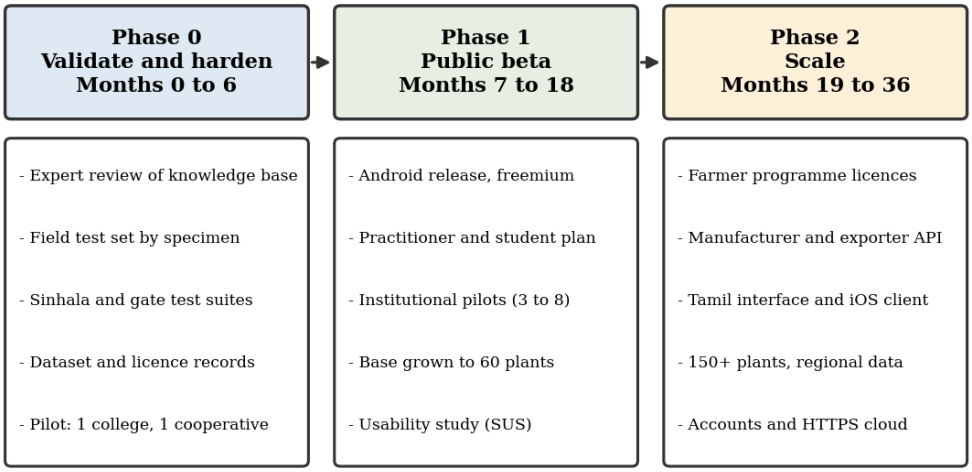


Figure 7.1 Commercialization roadmap

Channels combine direct and partner routes. Ayurveda colleges and clinics give access to practitioners and students and provide expert reviewers. Cooperatives and extension program reach farmers who may not install apps independently and can use assisted onboarding. Incubators and innovation grants provide early funding and mentoring. Word of mouth within practitioner communities is expected to matter more than paid advertising, which keeps acquisition cost low.

7.3.4 Cost structure and financial projection

The projection below is a planning model for three years. Its assumptions are listed in Table 7.7. We use an exchange rate of about LKR 320 per US dollar for foreign-priced services. At the published launch prices of the answer model [19], one grounded answer of about 1,500 input and 400 output tokens costs a small fraction of a rupee, so model usage is a minor part of cost compared with people and content review. Prices can change, and a fallback to a self-hosted open model is part of the risk plan.

Table 7.7 Key assumptions (to be tested in pilots)

Assumption	Year 1	Year 2	Year 3
Paying Plus subscribers (average in year)	120	450	1,200
Institutional licenses	3	8	18
Farmer-program licenses	2	6	14
Average revenue per Plus subscriber-year (LKR)	6,000	6,000	6,000
Grant and pilot funding (LKR)	1,200,000	1,500,000	1,000,000

Table 7.8 Three-year financial projection (LKR)

Item	Year 1	Year 2	Year 3
Plus subscriptions	720,000	2,700,000	7,200,000
Institutional licences	540,000	1,440,000	3,240,000
Farmer-programme licences	500,000	1,500,000	3,500,000
Grants and pilot funding	1,200,000	1,500,000	1,000,000
Total revenue and grants	**2,960,000**	**7,140,000**	**14,940,000**
Staff and contractors	2,400,000	4,800,000	7,200,000
Cloud and model usage	600,000	1,000,000	1,800,000
Content curation and expert review	600,000	1,000,000	1,500,000
Marketing and field pilots	400,000	900,000	1,400,000
Legal, IP and compliance	300,000	300,000	400,000
Administration	200,000	400,000	700,000
Total operating cost	**4,500,000**	**8,400,000**	**13,000,000**
Net result	**-1,540,000**	**-1,260,000**	**1,940,000**
Cumulative position	**-1,540,000**	**-2,800,000**	**-860,000**

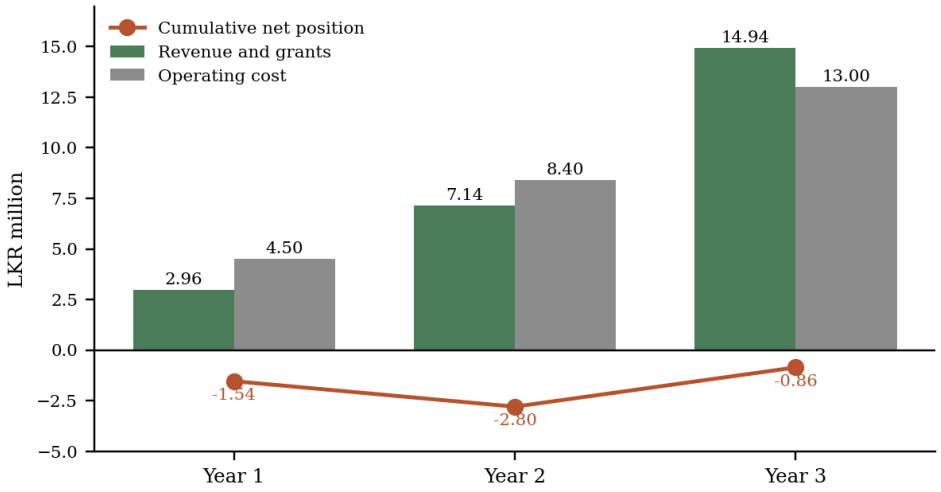


Figure 7.2 Projected revenue, cost and cumulative position

On these assumptions the venture runs at a loss in Years 1 and 2, reaches an operating surplus of about LKR 1.94 million in Year 3, and has a peak cumulative deficit of about LKR 2.80 million at the end of Year 2. Cumulative break-even therefore falls in Year 4. A seed funding target of about LKR 4 million, including a contingency, would cover the deficit period. The projection is sensitive to the pace of institutional adoption, so the pilots in Phase 0 are designed to measure willingness to pay before further investment.

7.3.5 Technical and operational feasibility

Technical feasibility is supported by a working prototype of all pipelines: the assistant retrieves and answers in both languages, the identification service classifies and routes images, and the cultivation interface and API are in place. Operational feasibility depends on four upgrades: cloud hosting with HTTPS and monitoring; authenticated account storage; a managed process for reviewing and updating knowledge entries; and a plan for offline classification, since exported lightweight model files exist but are not yet used on the device. A small team of two developers, one content and partnerships lead and part-time expert advisers is sufficient for Phase 1.

7.3.6 Risks and mitigation

Table 7.9 Principal risks and mitigation

Risk	Likelihood	Impact	Mitigation
Unsafe or wrong health information	Medium	High	Evidence gate, mandatory precautions, expert review of entries, disclaimer, incident log
Poor Sinhala answer quality	Medium	High	Native-speaker review, plant-name alias list, lower generation penalties, separate Sinhala test suite
Misidentification of a species, including toxic look-alikes	Medium	High	Confidence warnings, unknown-input rejection, expert-review route, restriction to supported classes
Vendor dependence and price change for language services	Medium	Medium	Abstraction layer, caching, evaluated open-model fallback
Insufficient or biased cultivation data	High	Medium	Partner data collection, regional datasets, advisory wording, field trials
Low willingness to pay	Medium	High	Free core tier, institution pilots, grant income, low fixed costs
Licence or data-provenance problems	Medium	High	Dataset provenance manifest, licence audit, replacement of restricted components before launch
Limited reach because many people are offline	High	Medium	Assisted onboarding through clinics and cooperatives; offline mode in Phase 2

7.3.7 Milestones and performance indicators

Table 7.10 Milestones and indicators

Phase	Milestone	Indicator and target
0	Knowledge base reviewed by practitioners	All 20 entries signed off; reviewer comments resolved
0	Field test set built	Specimen-separated set with reviewed labels;

		accuracy reported on it
0	Pilots completed	1 college, 1 cooperative; SUS score and task completion recorded
1	Public beta live	Android release; monthly active users and retention tracked
1	Paid adoption	Targets in Table 7.7 met for Plus subscribers and institutions
2	Scale	Base of 150 or more plants; farmer programmes active; operating surplus

7.3.8 Scenario and sensitivity analysis

Because the projection depends on adoption, we tested three scenarios for Year 3. The conservative case assumes 60 per cent of the planned volumes and the optimistic case 140 per cent. Staff, legal and administration costs stay fixed in all cases, while cloud, content and marketing costs scale with volume.

Table 7.11 Year 3 scenarios (LKR)

Scenario	Plus subscribers	Institutions	Farmer programmes	Revenue and grants	Operating cost	Net result
Conservative (60%)	720	11	8	9,300,000	11,120,000	-1,820,000
Base (100%)	1,200	18	14	14,940,000	13,000,000	1,940,000
Optimistic (140%)	1,680	25	20	20,580,000	14,880,000	5,700,000

The base and optimistic cases give an operating surplus in Year 3, while the conservative case gives a deficit. The result shows that institutional and farmer-programme licences, which together account for most of the revenue, decide viability, and that the pilots must secure committed institutional customers before large hiring. If volumes reach only the conservative level, hiring is slowed and the plan relies longer on grant income.

7.3.9 Unit economics

The cost of answering one question is small compared with the price of a plan. Using the planning figures stated earlier, one grounded answer costs about LKR 0.15 in model usage. Table 7.12 compares the variable cost per user with the price.

Table 7.12 Variable cost per user (planning assumptions)

Item	Community user	Plus subscriber
Answers per month	20	300
Model cost per month (LKR)	3	45
Hosting and storage allocation per month (LKR)	10	40
Revenue per month (LKR)	0	500 (annual plan average)
Contribution per month (LKR)	-13	about 415

A Community user costs about LKR 13 a month, so the free tier is affordable if one in about thirty free users becomes a paying subscriber, or if partner funding covers the free tier in public-interest programmes. A Plus subscriber contributes about LKR 415 a month before staff and overheads, which are covered at the subscriber volumes in Table 7.7.

7.3.10 Marketing and communication plan

Table 7.13 Marketing activities

Activity	Audience	Timing	Share of Year 1 marketing budget
Demonstrations at Ayurveda colleges and clinics	Students, practitioners	Phase 0 and 1	35%
Short Sinhala video guides on plant identification and safe use	General public	Phase 1	20%
Cooperative and extension workshops with printed quick guides	Farmers	Phase 1	25%
Presence at innovation and agriculture events	Investors, institutions, manufacturers	Phase 1	10%
Referral rewards for practitioners and students	Plus prospects	Phase 1	10%

Messages stress safety, local language and expert review rather than novelty. Every public claim will be checked against the safety wording in Section 7.4.4.

7.3.11 Team and operating plan

Table 7.14 Staffing plan (full-time equivalents)

Role	Year 1	Year 2	Year 3
Machine learning and backend engineering	1.0	1.0	1.5
Mobile and platform engineering	0.5	1.0	1.0
Product, partnerships and operations	0.5	0.5	1.0
Content curation (supported by part-time Ayurveda advisers)	0.0	0.5	0.5
Total full-time equivalents	2.0	3.0	4.0

Average annual cost per FTE (LKR)	1,200,000	1,600,000	1,800,000

These totals match the staff and contractor line in Table 7.8. Ayurveda advisers and botanical reviewers are engaged on honoraria within the content curation line, so that expert review remains independent of the development team.

7.4 Adoption intellectual property and compliance

7.4.1 Adoption barriers

The main barriers are trust, language quality, digital access and habit. Practitioners will adopt only a tool that experts have reviewed, so endorsement by an Ayurveda institution is a precondition for scale. Sinhala answers must be consistently natural before they are promoted. Farmers may need assisted onboarding and low-data use. We address these through expert review, native-speaker testing, partner-led training and a free tier.

7.4.2 Ownership, licensing and intellectual property

Ownership of the code, models and curated content must be confirmed under the institute's intellectual property policy and recorded in a written agreement among the four members before commercial use. Copyright in software and in the compiled knowledge base is protected under Sri Lanka's Intellectual Property Act [23], and we expect to rely on copyright, trademark registration of the brand and trade secrets for the curated data rather than on patents. Three licensing actions are required before launch. The permissions and provenance of the image datasets must be documented, because the source of the herb image collection has no confirmed reuse licence. The licence of the object-detection library used for organ detection must be reviewed, since its open-source licence carries obligations for network services and a commercial licence or an alternative detector may be needed [20]. The licences of all other open-source dependencies must be inventoried and honoured. Knowledge entries compiled from published literature will be attributed, and traditional knowledge will be treated respectfully, with practitioners acknowledged as contributors and benefit sharing discussed in partner agreements.

7.4.3 Data protection and security

The assistant collects no personal data by design. Introducing accounts, subscriptions, saved histories and expert-review images, as the commercial plan requires, brings the product within the Personal Data Protection Act [17]. We will collect the minimum data, state the purpose, obtain consent for image sharing and model-improvement use, set retention limits, and strip image metadata, which the review workflow already does. Security controls follow OWASP guidance for language-model applications [16]: input validation and bounds, protection against prompt injection, restricted cross-origin settings, protected expert credentials, rate limiting and HTTPS.

7.4.4 Regulatory position and professional responsibility

The practice and registration of Ayurveda practitioners are regulated by the Ayurveda Act [24], and medicines and medical devices by the National Medicines Regulatory Authority Act [25]. HerbIntel Lanka is positioned as an educational and decision-support tool. It does not diagnose, prescribe or give personalised doses, and its public wording will avoid therapeutic claims. We will seek written legal advice on whether any feature, particularly condition assessment, could be treated as a regulated product, and will keep all marketing consistent with that advice. Terms of use will state that the service is not a substitute for a practitioner or agronomist and that a cultivation recommendation does not guarantee a result.

7.4.5 Ethics and responsible AI

We commit to transparency about uncertainty, refusal when evidence is weak, human expert review of knowledge, disclosure of AI use, ongoing bias checks on datasets that may favour some regions or plants, and an accessible route for users to report errors. Incidents involving unsafe output will be logged and reviewed by the content team.

7.5 Sustainable Development Goal contribution

Table 7.15 Contribution to SDG targets and sustaining measures

Goal and target	Plausible contribution	Possible negative effect	Measure to sustain benefit
SDG 3 Good Health and Well-being	Safer use of traditional remedies through	Over-reliance on the app in place of	Disclaimers, referral wording, practitioner

	precautions and evidence levels (indirect contribution)	professional care	review
SDG 4, target 4.7 (education for sustainable development and appreciation of cultural diversity)	Bilingual, searchable record of traditional knowledge for students and the public	Errors in educational content	Expert sign-off, error reporting, version history
SDG 8, target 8.3 (support for productive activities and small enterprises)	Better crop choice and quality checks for small farmers	Recommendations that fail because of local conditions or prices	Regional data, advisory wording, field trials, partner support
SDG 12, target 12.2 (sustainable management of natural resources)	More efficient use of land and inputs through informed selection	Monoculture pressure if few plants are recommended	Broaden the plant base, diversity-aware recommendations
SDG 15, target 15.5 (protect natural habitats and biodiversity)	Cultivation guidance may reduce collection from the wild (potential, not measured)	Increased collection of popular species	Cultivation-first messaging, conservation notes in entries

We make no claim that these effects have been measured. To sustain them we will track indicators such as the number of farmers onboarded, the share of recommendations followed, user-reported safety incidents, and the growth and review status of the knowledge base, and we will report them to partners annually.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

The common problem was how to give users and farmers one reliable and accessible route from an unknown Sri Lankan medicinal plant to trustworthy knowledge and to an informed cultivation decision. Our integrated mobile ecosystem answers this problem with four connected components that share one Flutter application and one Flask backend.

The Herbal Knowledge Intelligence System gives grounded answers in English and Sinhala from a curated 20-plant knowledge base. It retrieved the correct plant entry for 5 of 6 English and 6 of 6 Sinhala questions, refused no answerable question, and routed all eight audience-interaction tests to the expected categories, with precautions and evidence levels stated in its answers. The plant identification component names plants from leaf, seed or spice and flower images using EfficientNetV2 classifiers and a YOLO11 organ detector, and recorded validation accuracies of 93.33% for leaf, 100% for seed or spice and 91.43% for flower. The quality assessment component identifies six supported species with 96.66% accuracy, classifies 36 leaf health conditions with 89.27% accuracy and leaf maturity with 91.74% accuracy, and uses Grad-CAM so that users can see why a condition was reported. The cultivation component delivers a validated input form and a recommendation service built on soil type, land size, budget and local weather.

Together these components cover the full path from recognising a plant, to understanding it safely, to deciding what to grow. Their narrow interfaces let each component work on its own and as part of the whole, and the safety features of the system, which are evidence gating, mandatory precautions, confidence warnings and visual explanation, apply across the entire journey. The commercialisation plan in Chapter 7 shows how the system can reach practitioners, students, farmers and institutions through a freemium model, with a three-year projection that reaches an operating surplus in the third year. We conclude that the integrated design meets its aim and provides a practical foundation for a Sri Lankan herbal knowledge and cultivation service.

8.2 Principal integrated contribution

Beyond the four components, the team contributes four things. First, an integration pattern of narrow label-in, result-out interfaces that lets independent research components operate as one product. Second, a safety-first design principle applied across components, in which uncertainty is always shown to the user through confidence warnings, evidence gating and refusal, mandatory precautions and visual explanation. Third, an evaluation approach and error taxonomy for bilingual herbal assistants, together with an audit method for image datasets that checks duplicates and provenance. Fourth, a complete commercialisation plan that links the technical work to validation, licensing, pricing and funding steps.

8.3 Future directions

The system was built so that another researcher or team can pick it up and extend it without starting again. Each component has a clear boundary, a narrow interface and its own data and evaluation scripts, so work can continue on one component at a time. The pathways below describe where a successor can begin, what to build on, and what the extension would deliver.

Table 8.1 Pathways for continuing the work

Area	Starting point for a successor	Suggested extension	Outcome
Knowledge assistant (C1)	Knowledge entries are plain text files that are re-ingested to rebuild the vector store; evaluation scripts and the question sets are reusable	Add plants with per-use evidence, separate plant and disease entries by metadata, add hybrid dense and keyword retrieval, and test lower generation penalties for Sinhala	A wider and more accurate bilingual assistant
Assistant evaluation	Paired Sinhala and English questions and the interaction suite already exist	Add questions outside the knowledge base, multi-plant fact-mixing tests, repeated runs and automated RAG evaluation [26], guided by known failure points [27]	Measured safety of the evidence gate
Identification (C2)	Training script, saved labels, validation reports and the expanded 25-class leaf model are available	Activate the larger leaf model with matching labels, collect specimen-separated field images, calibrate scores and add unknown-species rejection	Higher and better-trusted coverage
Mobile use (C2)	Lightweight model files are already exported	Run classification on the device with a mobile interpreter	Offline identification

Quality assessment (C3)	Three-stage pipeline, dataset manifests, benchmark logs and CSV knowledge files for diseases and maturity	Add more species and conditions, re-partition the data by subject, and pass the Grad-CAM evidence into the assistant answer	Wider and cleaner explainable assessment
Cultivation (C4)	Input form, validation, API and dataset structure are in place	Enlarge the dataset by region, add verified market prices, rainfall and seasonal data, try ensemble and boosting models, and explain each recommendation	More dependable recommendations
Integration	Narrow label-in, result-out contracts (Appendix A)	Pass the identified plant name automatically to the assistant and the cultivation module, and record a full multi-device run	One continuous user journey
Product and operation	Production steps are described in Chapter 5 and the plan in Chapter 7	Move to cloud hosting with HTTPS and accounts, add a Tamil interface and voice input, and run the Phase 0 pilots	A deployable service

A successor should work in this order: first review the knowledge and image data with domain experts, then extend the models and datasets, then complete the integration and operation steps, and last follow the commercialisation roadmap in Chapter 7. Each step builds on the one before and uses the artefacts already in the repository.

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APPENDICES

Appendix A Integrated interface specification

Table A.1 lists the interfaces on which the four components depend. Field names follow the implemented contracts.

Table A.1 Interface specification

Endpoint	Method	Request fields	Response and errors
/ask	POST	question; session identifier; user context with home category (default_type) and optional explicit category (user_type)	answer; sources; categories used; safety_context_available; evidence_references_available
/api/predict	POST	images (repeated multipart field); model_type (plant, seed, flower); optional validate_category	label, category, scores, alternatives, quality feedback, plant information; 400 invalid input; 503 model unavailable; category_mismatch error; warning below 0.95
/api/detect-organs	POST	single image; optional confidence	normalised boxes and scores; 400 or 503
/api/identify-live	POST	single image; optional confidence	classification of the strongest detection; null identification if none
/api/verifications	POST	images; AI label; AI confidence; consent	201 pending request; 400 invalid input
/api/verifications/{id}	GET	request identifier	public status; 404 unknown identifier
/api/expert/verifications/...	GET, POST	expert key header	queue and review; 401 unauthorised
/api/health	GET	none	dependency and artefact status
/api/quality/validate-image	POST	one image	validity with reason code; 422 if image is too small, blurred or badly lit
/api/quality/identify-plant-multiple	POST	one to three images	species with confidence; 422 for unsupported or rejected input
/api/quality/assess-condition	POST	one to three images; optional manual_inputs JSON (leaf length and width)	species, condition, maturity, suitability, Grad-CAM image (Base64)
Cultivation recommendation route	POST	soil type; land size; budget in LKR; location	recommended herbs; validation error for invalid input
All routes	any	request larger than 25 MiB	413 JSON error; 404 and 405 JSON errors

Appendix B Team contribution declaration

The table records the agreed individual and shared contributions. All four members confirm it by signing below.

Table B.1 Contribution declaration

Member and ID	Component	Individual contribution	Shared contribution	Signature
Munasinghe M P, IT22640598	C1 Knowledge assistant	Knowledge base, RAG engine, gate, audience resolution, chat client, evaluation	Shared application shell, disease entries used by C3	
H D Sachintha, IT22168986	C2 Identification	Classifiers, preprocessing, organ detector, review backend, identification page	Shared application shell and backend routes	
Pattiya Arachchi P A H K, IT22112132	C3 Quality assessment	Three-stage EfficientNetV2-B0 pipeline, image validator, Grad-CAM, knowledge lookups, quality page	Species and condition context to C1; shared application shell	
Wickramasinghe D G T K, IT22132796	C4 Cultivation	Dataset, models, recommendation API, input form	Shared application shell and backend routes	

Appendix C Glossary

Table C.1 explains terms that are used across the components.

Table C.1 Glossary of terms

Term	Meaning in this report
Evidence gate	A rule in the knowledge assistant that discards retrieved text beyond a set distance from the question, so that answers use only closely related entries
Grad-CAM	A method that shows which image regions most influenced a prediction by overlaying a heat map
Multi-view averaging	Averaging the model probabilities of several views or images of the same sample to give a steadier result
Unknown class	An explicit class for inputs that are not one of the supported plants, used to stop further analysis
Species-conditioned head	A separate output layer for each species on a shared model, so that each species uses only its own condition classes
Audience category	The user type (general public, practitioner, student or researcher) that sets the depth and tone of an answer
Embedding	A numeric representation of text used to find knowledge entries similar to a question
Freemium	A pricing model with a free core service and paid plans for deeper or

	larger use
Narrow interface	A contract in which a component accepts a small input, such as a label, and returns a structured result

Appendix D Deployment and configuration summary

Table D.1 summarises the settings needed to run the prototype and the changes planned for production.

Table D.1 Deployment and configuration summary

Item	Prototype setting	Planned production setting
Backend service	Flask on port 5000, local server	Containerised service behind HTTPS on a cloud host
Language service key	Environment variable	Managed secret store
Vector store	Persisted local ChromaDB, rebuilt from knowledge files when needed	Managed persistent volume with scheduled backup
Model files	Loaded lazily on first request and cached	Versioned artefacts with health checks
Upload limit	25 MiB per request with JSON error bodies	Same limit with rate limiting and monitoring
User data	No personal data collected	Authenticated account store with retention rules
Expert review queue	SQLite with key-protected endpoints	Managed database with role-based access
Cross-origin policy	Open for development	Restricted to the application origins
Client	Flutter application connected to the configured server address	Signed Android release, iOS build in Phase 2